MIL-L-13802(Cm10)
22 November 1954

MILITARY SPECIFICATION
LACQUER, CHLORINE RESISTANT; FOR FIBER CONTAINERS

1. SCOPE

1.1 This specification covers a lacquer for use on interior surfaces of containers used in packing impregnites, C02 and XXC03.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT-P-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

MILITARY

JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
MIL-B-107 - Boxes, Wood, Wirebound (Overseas Type).
MIL-C-124 - Containers (Cans, Pails and Drums) Metal (For Other Than Subsistence Items.)
MIL-I-292 - Impregnite, XXC03.

U. S. ARMY

97-54-301 - Drum, Fiber, 20-Gallon Capacity.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification.- The lacquer shall be a product which has been tested and has passed the qualification tests specified in 4.4.1 (see 6.5).

3.2 Material.

3.2.1 Toxicity.- The lacquer shall contain no chlorinated compounds or hydrolyzable chlorine derivatives. The contractor shall certify that this requirement has been complied with.

3.2.2 Dilution stability.- There shall be no curdling, precipitation, or separation after standing 4 hours when tested as specified in 4.4.3.1.

3.3 Dried film characteristics.

3.3.1 Drying time.- The material shall reach the tack-free, print-free, and full-hardness states in time intervals not exceeding 3, 10, and 60 minutes, respectively, when tested as specified in 4.4.4.1.

3.3.2 Dry opacity.- The contrast ratio shall be 0.98 (min.). The wet film thickness shall not exceed 0.002 inch when tested in accordance with Method 412.1 of Specification TT-P-141.

3.3.3 Flexibility.- The material shall not crack or flake when tested as specified in 4.4.4.2.

3.3.4 Specular gloss.- The 60-degree specular gloss shall be 20 ± 2°, when tested in accordance with Method 610.1 of Specification TT-P-141.

3.3.5 Knife test.- The material shall adhere tightly and shall not flake, crack, or powder from the panel, and the cut shall show beveled edges, when tested as specified in 4.4.4.3.

3.3.6 Heat stability.- The material shall satisfy the toughness and specular gloss requirements (q.v.), when tested as specified in 4.4.4.4.

3.3.7 Recoating.- The material shall not be self lifting, nor show evidence of film irregularity when tested as specified in 4.4.4.5.

3.3.8 Impregnite resistance.- The lacquer [illegible] no deterioration such as blistering, cracking, peeling, softening, [illegible] of failure due to the presence of impregnite when tested as specified in 4.4.4.6.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.- A lot shall consist of the material produced by one manufacturer in not more than 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials, providing the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2 Sampling.- A representative 1-quart specimen shall be removed from each of five representative containers taken from each lot by the Government inspector. Each specimen shall be placed in a separate clean, dry container labeled to identify the specimen with the lot and container represented. Each specimen shall be tested separately as specified in 4.4. If there are fewer than five containers in the lot, a specimen shall be removed from each container.

4.3 Inspection.- The inspector shall inspect the packaging, packing, and marking for compliance with section 5.

4.4 Tests

4.4.1 Qualification tests.- Qualification tests shall consist of all tests specified herein.

4.4.2 Inspection tests.- Inspection tests for acceptance of individual lots shall consist of all tests with the exception of 4.4.4.6.

4.4.3 Material.

4.4.3.1 Dilution stability.- The dilution stability of the lacquer shall be determined as specified in Method 439.1 of Specification TT-P-141. The diluent shall be composed of ethyl alcohol and toluene (10 and 90 percent by volume, respectively), and the lacquer shall be diluted in the ratio of 15 parts of diluent to 85 parts of lacquer by volume. Observe for compliance with the dilution stability requirement of 3.2.2.

4.4.4 Dried film characteristics.

4.4.4.1 Drying time.

4.4.4.1.1 Tack-free.- The time required for the lacquer to reach the tack-free state shall be determined in accordance with Method 406.1 of Specification TT-P-141, except that the lacquer shall be applied to the test panel in accordance with Method 207.1 of Specification TT-P-141, to yield a wet film thickness of 0.004 inch. The result shall be checked for compliance with the drying-time requirement of 3.3.1.

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4.4.4.1.2 Print-free.- The time required for the lacquer to reach the print-free state shall be determined in accordance with Method 621.1 of Specification TT-P-141, except that no felt shall be interposed between the weight and the cheese cloth. The test panel shall be prepared in accordance with Method 406.1 of Specification TT-P-141. The pressure shall be at least 1 p.s.i., and the temperature during the test shall be 25 ± 1°C. After the lacquer has been allowed to dry for 10 minutes, pressure shall be immediately applied for 20 minutes, and the lacquer shall then be examined for compliance with the drying-time requirement of 3.3.1.

4.4.4.1.3 Full-hardness.- The time required for the lacquer to reach the full-hardness state shall be determined when the lacquer is applied as specified in 4.4.4.1.1, except that the wet-film thickness shall be at least 0.005 inch on a steel panel which has been phosphate-cleaned in accordance with Method 201.1 of Specification TT-P-141. Check for compliance with the drying-time requirement of 3.3.1.

4.4.4.2 Flexibility.- The flexibility of the lacquer shall be determined in accordance with Method 622.1 of Specification TT-P-141. The lacquer shall be stirred thoroughly and applied to yield a dry-film thickness of 0.0010 ± 0.0001 in. and the panel shall be cured by baking at a temperature of 105 ± 2°C. for the length of time specified by the manufacturer and conditioned for 1/2 hour at 25 ± 1°C. The mandrel diameter shall not exceed 1 inch. Observe for compliance with the flexibility requirement of 3.3.3.

4.4.4.3 Knife test.- The toughness of the lacquer shall be determined in accordance with Method 617.1 of Specification TT-P-141. The panel shall be prepared as specified in 4.4.4.2, and checked for compliance with the toughness requirement of 3.3.5.

4.4.4.4 Heat stability.- Place 7 ounces of the lacquer in an 8-ounce glass jar, close the jar tightly, and immerse the lower half of the jar in water maintained at 76 ± 2°C. After 24 hours remove the jar from the water and allow the contents to return to 25 ± 1°C. Determine the toughness and specular gloss of the lacquer as specified in 4.4.4.3 and Method 610.1 of Specification TT-P-141, respectively, and check for compliance with the heat-stability requirement of 3.3.6.

4.4.4.5 Recoating.- Prepare four test panels as specified in 4.4.4.2, but air-dry them for periods of 4, 8, 24, and 48 hours, respectively. Immediately after each panel has dried for its prescribed period, immerse it to a depth of 2-1/2 inches for at least 5 seconds in lacquer which has been diluted as specified in 4.4.3.1. As each panel is removed from the lacquer, orient it vertically, allow to dry, and observe for compliance with the recoating requirement of 3.3.7.

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4.4.4.6 Impregnite resistance.--Apply the lacquer in a manner specified by the manufacturer to the inside of three fiber drums conforming to Specification 97-54-301. Fill the drums with impregnite XXCC3 and package as specified in Specification MIL-I-292. Place the drums in tropical storage (113°F. and 85-percent relative humidity) for 6 months. Examine for compliance with paragraph 3.3.8.

4.5 Rejection and resubmission.- If any specimen fails to meet any requirement of this specification, the lot represented shall be rejected. The contractor at no expense to the Government shall have the option of having a partial or complete analysis made of each container in the rejected lot. He may then remove the nonconforming containers and resubmit the remaining portion of the lot for acceptance testing. If any specimen from the resubmitted lot fails to conform to this specification, the lot shall be finally rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Lacquer shall be furnished in 1-gallon, multiple-friction cans or in 5-gallon, lug-covered steel pails, as specified in the contract or order (see 6.2).

5.1.1 One-gallon can.- One-gallon, multiple-friction cans shall be made of special coated manufacture's terneplate and shall conform to the requirements of Specification MIL-C-124, except that no exterior protective coating will be required.

5.1.2 Five-gallon pail.- Five-gallon, lug-cover steel pails shall conform to the requirements of Specification MIL-C-124.

5.2 Packing

5.2.1 One-gallon cans.- Four 1-gallon cans shall be packed in a nailed wood or wire-bound box conforming to the requirements of Specification JAN-P-106 or MIL-B-107, respectively.

5.2.2 Five-gallon pail.- Five-gallon lug-cover steel pails will require no further packing.

5.3 Marking.- In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The lacquer covered by this specification is intended for use as a finish coat on the interior surfaces of containers used for the packing of impregnites CC2 and XXCC3.

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6.2 Ordering data.- Procurement documents should specify:

a. The title, number, and date of this specification.

b. Size of container (see 5.1).

6.3 Batch.- A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.4 Sampling and testing.- When the contractor consistently produces high quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.5 Qualification.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Commanding Officer of the Chemical Procurement District in which the supplier is located.

Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

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Other interests:

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